

Difference between TCP and UDP

Transmission Control Protocol (TCP) and **User Datagram Protocol** (UDP) are the most widely used **Internet protocols**. TCP is a connection-oriented protocol, which means, once a connection is established, data can be sent bidirectional. UDP, on the other hand, is a simpler, connectionless Internet protocol. Multiple messages are sent as packets in chunks using UDP.

Go through this article to find out more about TCP and UDP and how they differ from each other.

What is TCP?

TCP is a **transport layer** connection-oriented protocol. It provides a reliable connection and protected data transmission between the connected machines over a network. It first establishes a secure link and then sends the data.

TCP sends the data from one device to the other in the form of data blocks. It is slow in data transmission but has more functionalities such as flow control, error control, and congestion control in the system.

The TCP header is of 2060 bytes and therefore, it contains several pieces of information to improve the reliability. But, the overhead is raised. Because of its reliability, protocols like HTTP, FTP, etc., use TCP for secured data transmission over the network.

What is UDP?

UDP is a transport layer, connectionless protocol. It provides quick transmission of data between the connected machines over a network. There is no overhead of creating, maintaining, and terminating a connection in UDP.

It is mainly used to transmit real-time data where we cannot afford any transmission delays. UDP sends the data from one device to the other in the form of continuous data streams.

The UDP header is of fixed size, i.e., 8 bytes. It is unreliable but faster in speed. Due to its transmission speed, protocols like DNS, DHCP, RIP, etc., use UDP for proper data transmission over the network.

Differences between TCP and UDP

The following table highlights the major differences between TCP and UDP.

Key	TCP	UDP
Definition	It is a communications protocol, using which the data is transmitted between systems over the network. In this, the data is transmitted in the form of packets. It includes error-checking, guarantees the delivery and preserves the order of the data packets.	It is same as the TCP protocol except this doesn't guarantee the error-checking and data recovery. If you use this protocol, the data will be sent continuously, irrespective of the issues in the receiving end.
Design	TCP is a connection-oriented protocol.	UDP is a connectionless protocol.
Reliability	TCP is more reliable as it provides error checking support and also guarantees delivery of data to the destination router.	UDP, on the other hand, provides only basic error checking support using checksum. So, the delivery of data to the destination cannot be guaranteed in UDP as in case of TCP.
Data transmission	In TCP, the data is transmitted in a particular sequence which means that packets arrive in-order at the receiver.	There is no sequencing of data in UDP in order to implement ordering it has to be managed by the application layer.
Performance	TCP is slower and less efficient in performance as compared to UDP. Also TCP is heavy-weight as compared to UDP.	UDP is faster and more efficient than TCP.
Retransmission	Retransmission of data packets is possible in TCP in case packet gets lost or needs to be resent.	Retransmission of packets is not possible in UDP.
Sequencing	The Transmission Control Protocol has a function that allows data to be sequenced (TCP). This implies	In UDP, there is no data sequencing. The application

	that packets arrive at the recipient in the sequence they were sent.	layer must control the order if it is necessary.
Header size	TCP uses a variable-length (20-60) bytes header.	UDP has a fixed-length header of 8 bytes.
Handshake	Handshakes such as SYN, ACK, and SYNACK are used.	It's a connectionless protocol, which means it doesn't require a handshake.
Broadcasting	Broadcasting is not supported by TCP.	Broadcasting is supported by UDP.
Examples	HTTP, HTTPs, FTP, SMTP, and Telnet use TCP.	DNS, DHCP, TFTP, SNMP, RIP, and VoIP use UDP.

Conclusion

From the above discussion, we can conclude that TCP is the preferred protocol where reliability of data weighs more than transmission speed. UDP is connectionless and its header size is lightweight, which is why it is fast, but not as reliable as TCP.

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